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ADDITIONAL DOCUMENT FOR PROJECT PROPOSAL

The Round Table Exchange (RTE): A Multi-Party Skill Barter Network with Graph-Based Cycle Matching

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Executive Summary

The Round Table Exchange (RTE) is a skill exchange system that automatically finds trade loops among three or more people without using money. Existing skill exchange platforms rely on direct 1-to-1 trades (where Person A trades directly with Person B) or time credits. The Round Table Exchange (RTE) solves this problem by representing users and skill offers as a graph, running a fast cycle-detection algorithm to find groups of 3 or more people who can satisfy each other's skill requests, and ranking trades by location and availability. The project includes a simple web app for skill listings, a multi-user trade confirmation workflow, and a synthetic data generator for testing. Trust scores, dispute resolution, and calendar sync are saved for future work.

Project Background

0.1 The Skill Barter Landscape

Skill bartering — exchanging services directly without money — has existed for a long time. Digital platforms today usually follow two main patterns:

Direct 1-to-1 swap platforms (e.g., Reciproc8, Ying, BarterQuest, The Barter Shop) require both users to want what the other person offers at the exact same time. This limits viable matches. For example, if Alice teaches Python and wants guitar lessons, she can only trade with someone who teaches guitar and wants Python. These platforms do not find multi-person trade chains.

Time-credit currency platforms (e.g., TimeRepublik, hOurworld) solve the matching problem by giving users time credits (1 hour of service = 1 credit). This turns bartering into a virtual credit system rather than a direct exchange.

0.2 Prior Art: Kidney Paired Donation Matching

Finding multi-person exchange loops in a graph is well known in medical kidney exchange programs [9, 3]. In kidney exchanges, incompatible donor-patient pairs are matched into exchange loops (Donor A gives to Patient B, Donor B gives to Patient C, and Donor C gives to Patient A). The Round Table Exchange (RTE) applies this same graph-matching idea to skill bartering.

0.3 The Problem Gap

Current skill barter platforms do not find multi-person trade loops automatically. Users on 1-to-1 platforms must search manually, while users on time-credit platforms give up direct barter. The Round Table Exchange (RTE) fills this gap by finding multi-person trade loops automatically while keeping direct barter semantics.

Problem Statement

Existing platforms cannot automatically find trade loops between three or more people. People looking to trade skills are forced to either find a direct 2-person partner — which gets very difficult for specific skills — or use virtual time credits. Because of this, many possible multi-person skill trades are missed simply because there is no tool to find them automatically.

Solution

The Round Table Exchange (RTE) solves the multi-party skill barter problem using three main features. Trust ratings, dispute management, and calendar scheduling are left for future work.

0.4 Matching Engine

The main core of The Round Table Exchange (RTE) is a **graph-based cycle matching engine**. Users and skill posts are stored as a directed graph. The engine runs a Depth-First Search (DFS) algorithm to find closed trade loops of 3 or 4 people. Found loops are scored and ranked based on location proximity and overlapping schedule times. A synthetic data generator is included to test the engine without needing real users.

0.5 Simple Application Layer

A simple web application provides user login, profiles, and skill offer/request listings (with tags, location, and time windows). An API connects the database to the matching engine.

0.6 Group Confirmation Workflow

When the matching engine finds a trade loop, all users in that loop get an in-app notification. An accept/reject workflow tracks responses: the trade is confirmed only if every user accepts before the time limit expires. If anyone rejects or the time expires, the trade proposal is canceled.

Formation Team

The Round Table Exchange (RTE) project is developed by a team of two students from the Department of Computer Science and Engineering, University of Information Technology and Sciences (UITS), Dhaka, Bangladesh.

Name	Student ID	Primary Technical Responsibilities
Ahmmad Abdali Khan	0432320005101118	Lead Software Engineer. Responsible for core graph matching engine development, bounded-depth DFS cycle discovery algorithm, ranking functions, PostgreSQL database schema design, REST API back-end services, and runtime complexity analysis.
Sumaia Bintey Ismail	0432320005101103	Application & Systems Engineer. Responsible for frontend web application development, skill listing management, synthetic data generator module, group confirmation workflow state machine, use-case specifications, and system documentation.

The responsibility distribution maps across the Work Breakdown Structure: Ahmmad Abdali Khan leads algorithm engineering, graph matching search, backend API infrastructure, and runtime complexity evaluation (WBS 3.1, 3.2, 3.3, 4.3, and 6.2), while Sumaia Bintey Ismail leads frontend development, skill listing user workflows, synthetic dataset generation, group confirmation state machine, use-case modeling, and project documentation (WBS 1.0, 2.0, 3.4, 3.5, 4.1, 4.2, 5.0, 6.1, and 7.0).

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